ARC2-Global: Tangent-Corrected Alignment for Embedding-Index Migration

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Abstract. Embedding-model upgrades must preserve retrieval quality while document re-encoding proceeds under a limited budget. We propose ARC2-Global, a compact document adapter that refines centered Procrustes alignment with two ridge-fitted tangent corrections. The updates target cosine-relevant directions and collapse into one affine map, preserving native destination queries without neural mapper training. Across four development collections, two encoder migrations and three anchor-inclusive budgets, ARC2-Global achieves the highest macro nDCG@10 among six evaluated methods. Its gain over Procrustes is 0.0243, with family-adjusted interval [0.0181, 0.0308]; the second correction adds 0.0069. On prospective TREC-COVID/E5, gains over the evaluated Converter range from 0.2000 to 0.2572. A descriptive comparison across all five collections further shows E5 ARC2 at 25% coverage exceeding Converter and affine ridge at 75%. On TREC-COVID/MiniLM, 75% coverage yields 0.5195 nDCG versus native Full-Y's 0.5203, with the highest Recall@100 point mean among the six adapters. The final 384-to-768-dimensional coefficient matrix occupies 1.13 MiB, using 114.7 times fewer parameter bytes than the evaluated Converter.

Keywords: Embedding migration · Vector retrieval · Representation alignment · Mixed galleries

1 Introduction

Embedding-model upgrades create a practical index-maintenance problem: destination-model queries must remain effective while legacy document vectors are replaced gradually. A staged migration learns from paired source and destination embeddings, maps the remaining documents, and replaces these proxies as true destination vectors become available. The useful operating point is an intermediate index that delivers strong retrieval quality with limited destination-document exposure.

The intermediate index must rank true destination embeddings and mapped proxies together. A mapper that works well on a proxy-only gallery can change behavior after true vectors are inserted. Cosine retrieval also depends on direction, whereas coordinate regression spends capacity on radial errors that disappear after normalization. This motivates a directional document adapter that preserves compact inference and searches with the native destination query.

Document-side conversion and query-side adaptation already provide routes across embedding spaces [17,11]. Orthogonal alignment supplies a compact geometric starting point, while local alignment and neural regression provide more flexible alternatives [16,2]. We investigate a complementary design: correcting the prediction error that changes vector direction while retaining one affine document map.

ARC2-Global applies two tangent corrections to a centered rectangular Procrustes map. For each paired document, it projects destination error onto the tangent plane at the predicted direction, rescales it to the raw prediction, and fits an affine ridge update. Repeating this operation twice yields coefficients that collapse into one affine map followed by normalization. Native destination queries then search true and proxy vectors together, with each newly encoded document replacing its proxy under the same identifier (Fig. 1).

The results identify a strong quality–exposure trade-off. On four development collections, the fixed 1,024-pair adapter leads all five baseline configurations in macro nDCG@10 across two migrations and three budgets, gaining 0.0243 over Procrustes. The second correction adds 0.0069 without adding an inference stage. On the prospective TREC-COVID collection, E5 gains over Converter and ridge persist at every budget. Across all five collections, E5 ARC2 at 25% coverage also exceeds these two alternatives at 75% in a descriptive comparison. Matched all-paid fitting strengthens MiniLM adaptation by another 0.0596 nDCG at the same paid-document budget.
Our contributions are:

- **Directional refinement with affine inference.** Two tangent-residual ridge updates target prediction direction and improve measured development retrieval while retaining one compact document map and native destination queries.
- **Strong quality at matched fitting exposure.** Across two migrations and three paid-document budgets, the four-collection development macro favors ARC2-Global over all five evaluated alternatives. Ablations quantify the value of the second update, and all-paid fitting tests the method beyond fixed-anchor supervision.
- **Useful quality at lower document exposure.** A held-collection evaluation confirms E5 gains over Converter and affine ridge at each tested budget. The descriptive 25%-versus-75% comparison exposes a useful migration operating point, while mixed-gallery diagnostics explain why adapter choice must be evaluated after native-vector insertion.

2 Related Work

Document-side translation and mixed databases. Embedding-Converter learns document-side conversion using regression together with global and local similarity losses [17]. Yang et al. study integration across embedding models through global alignment and locally fitted isometries, giving the A2M and LA2M approaches [16]. Their authors' repository also provides the SimpleLinear neu-

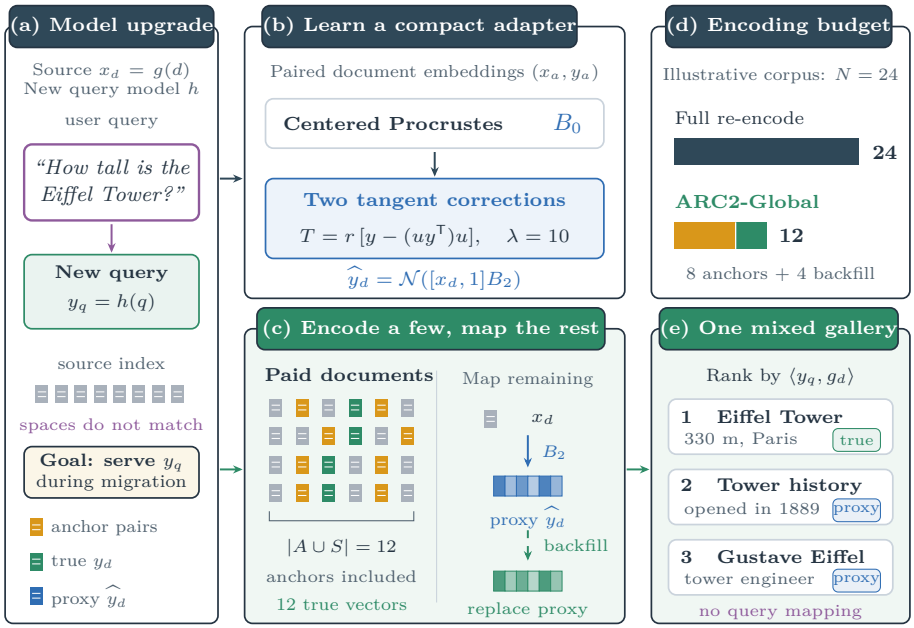


Fig. 1. A worked ARC2-Global migration. Paired document vectors train centered Procrustes and two tangent corrections, yielding one affine adapter. Unencoded documents remain searchable through proxies; backfill replaces them with native vectors in the same gallery. The 24-document corpus, eight anchors, four additional encodings and displayed hits are illustrative, not measured retrieval results.

ral mapper used in our comparison [2]. More recently, Yang and Cao study out-of-distribution translation, multi-model mixing, and chaining, with a hierarchical mixture of experts and residual regularization along target principal directions [15]. Their localized expert structure addresses broader multi-model integration; our directional target uses the tangent plane at each anchor's current prediction, and both corrections collapse into one global affine map. HMoE provides related-work context; our evaluated panel comprises the five configurations specified in Sect. 4. FastFill already combines visual feature alignment with uncertainty-based partial backfilling, and shows that replacement can initially reduce retrieval performance [3]. Building on these migration settings, we study the two-correction affine computation and its relevance-judged text-retrieval behavior under anchor-inclusive encoding budgets.

Compatibility and translation direction. Backward-compatible representation learning trains a new encoder to remain usable with previously computed gallery features [9]. Drift-Adapter instead fits a post-hoc map from new-model queries to an existing legacy index, comparing Procrustes, low-rank affine, and residual neural adapters [11]. Spherical interpolation refines Procrustes-aligned queries

through SLERP between old and aligned new query directions [7]; ARC2 instead fits tangent ridge updates to document anchors. ARC2-Global shares the use of paired embeddings and frozen encoders, but maps legacy document vectors into the destination space. Its evaluation concerns mapped and native destination documents competing as re-encoding progresses.

Geometric alignment and refinement. Orthogonal Procrustes provides the classical least-squares alignment underlying our initialization [8]. Recent work connects approximate preservation of pairwise inner products to bounds on Procrustes alignment error and demonstrates interoperability across retrieval models [5]. Unpaired translation pursues a different supervision regime: vec2vec learns translation through a shared latent representation without known correspondences [4], while mini-vec2vec combines pseudo-pair construction, Procrustes estimation, and iterative matching and clustering refinements [1]. Thus, linear translation, Procrustes initialization, and refinement are all established ingredients. ARC2-Global uses known document correspondences and a specific finite refinement: exactly two ridge-fitted tangent corrections to a centered rectangular Procrustes initialization. Both corrections are folded into one affine coefficient matrix, followed by row normalization at application time. This differs from discovering pseudo-correspondences, learning a latent translator, or routing documents among local maps.

Retrieval evaluation. BEIR emphasizes retrieval across heterogeneous tasks and domains [10], while NanoBEIR provides smaller collections suitable for development comparisons [18]. Sentence-BERT, MiniLM, and E5 supply established encoder families and training approaches relevant to our model transitions [6,14,13]. Our evaluation targets judged retrieval quality in actual mixed galleries, separating that quantity from agreement with a destination ranking or approximate nearest-neighbor overlap. This distinction matters because accurately reconstructing individual vectors and preserving their internal neighborhoods do not directly measure the relevance of the final combined ranking. The experimental question is therefore whether the correction improves retrieval under a matched encoding budget, and in which encoder and gallery conditions that improvement persists.

3 ARC2-Global: Angular Refinement for Mixed Galleries

ARC2-Global converts legacy document vectors into destination-space proxies and combines them with true destination vectors. Two direction-focused corrections refine one global affine map.

3.1 Migration Setting and Budget

Let a corpus contain N documents with stored, unit-normalized source embeddings $x_i \in \mathbb{R}^{d_s}$, written as row vectors. The destination encoder produces unit

vectors $y_i \in \mathbb{R}^{d_t}$, but obtaining every y_i requires encoding the entire corpus again. We seek destination-space document representations using only a budgeted set of destination document vectors. The dimensions may differ; our evaluated encoder pairs map $d_s = 384$ to $d_t = 768$. Queries are encoded directly by the destination query encoder; the document map aligns with that existing destination query space.

Let A be the paired anchor set used for fitting and S the set selected for document replacement. The unique destination-document budget is

$$b = |A \cup S|, \qquad b = \lfloor \rho N \rfloor. \tag{1}$$

Anchors consume this budget. Our selection starts with A and fills the remaining slots from a fixed random document order, skipping duplicates, so $A \subseteq S$ and $|S| = b$. Larger budgets extend the same order and retain the same fitted anchors. The principal setting uses $m = |A| = 1{,}024$ pairs; an all-paid-pairs comparison separately fits on every paid pair. Fit uses paired document vectors without queries or relevance labels. This separates the amount of fitting supervision from the number of true vectors in the gallery. The same map serves several replacement budgets; after full replacement, scores depend entirely on the destination encoder.

3.2 Centered Rectangular Initialization

A centered Procrustes map supplies a geometric starting point before angular refinement. Stack the source and destination anchors as $X_A \in \mathbb{R}^{m \times d_s}$ and $Y_A \in \mathbb{R}^{m \times d_t}$, with row means μ_s and μ_t. Using a thin singular value decomposition, compute

$$(Y_A - \mathbf{1}\mu_t)^\top (X_A - \mathbf{1}\mu_s) = U \Sigma V^\top, \qquad W = UV^\top. \tag{2}$$

Here $W \in \mathbb{R}^{d_t \times d_s}$ is rectangular and semiorthogonal. Centering permits translation; the rectangular map accommodates unequal dimensions. Its raw prediction is $(x - \mu_s)W^\top + \mu_t$.

We expand this prediction into an affine coefficient matrix so that subsequent corrections can be accumulated without adding inference stages. With $p = d_s + 1$ and $\widetilde{x} = [x, 1]$, define

$$B_0 = \begin{bmatrix} W^\top \\ \mu_t - \mu_s W^\top \end{bmatrix} \in \mathbb{R}^{p \times d_t}, \qquad \widetilde{x} B_0 = (x - \mu_s)W^\top + \mu_t. \tag{3}$$

This expansion preserves raw prediction norms, without row normalization or a fitted scale; those norms enter the correction targets. The raw affine expansion uses float64 coefficients computed from the float32 Procrustes means and matrix.

3.3 Two Tangent-Residual Ridge Corrections

Write $\mathcal{N}(z) = z/\|z\|_2$ for a nonzero vector z. Cosine retrieval depends on document direction, whereas a raw vector residual also contains a radial component.

We therefore project each anchor target onto the tangent plane at its current predicted direction. Let $\widetilde{X} = [X_A, \mathbf{1}]$. At update $t \in \{0, 1\}$, for anchor j set

$$z_j^{(t)} = \widetilde{x}_j B_t, \qquad r_j^{(t)} = \|z_j^{(t)}\|_2, \qquad u_j^{(t)} = z_j^{(t)}/r_j^{(t)},$$
$$T_j^{(t)} = r_j^{(t)} \left[y_j - \left(u_j^{(t)} y_j^\top\right) u_j^{(t)} \right]. \tag{4}$$

The bracketed term is orthogonal to $u_j^{(t)}$. The norm factor converts this directional discrepancy into a target in raw prediction coordinates.

The target has a direct cosine-loss interpretation. For a nonzero row prediction z and unit destination anchor y_j, define $\ell_j(z) = 1 - y_j \mathcal{N}(z)^\top$. Differentiation gives

$$\nabla_z \ell_j(z) = -\frac{y_j - (u y_j^\top) u}{\|z\|_2}, \qquad T_j^{(t)} = -(r_j^{(t)})^2 \nabla_z \ell_j(z_j^{(t)}). \tag{5}$$

The target is a norm-scaled negative cosine gradient. Shared affine regression smooths targets across anchors, so fitted row corrections need not be tangent.

One global ridge regression shares these correction targets across anchors. Stacking their rows into $T^{(t)}$, we solve

$$\Delta_t = \underset{\Delta \in \mathbb{R}^{p \times d_t}}{\arg\min} \|\widetilde{X}\Delta - T^{(t)}\|_F^2 + \lambda \|\Delta\|_F^2, \tag{6}$$
$$= \widetilde{X}^\top (\widetilde{X}\widetilde{X}^\top + \lambda I_m)^{-1} T^{(t)}, \qquad B_{t+1} = B_t + \Delta_t. \tag{7}$$

The penalty regularizes the correction, including its intercept. Equivalently, changing variables to the updated coefficient matrix yields

$$B_{t+1} = \underset{C}{\arg\min} \|\widetilde{X}C - (\widetilde{X}B_t + T^{(t)})\|_F^2 + \lambda \|C - B_t\|_F^2. \tag{8}$$

This fit centers its correction penalty on the current affine map while matching norm-scaled cosine descent targets. Unlike direct zero-centered ridge onto Y_A, the update retains the existing alignment while fitting its directional correction. We fix $\lambda = 10$ and perform exactly two updates, recomputing targets after the first; the development ablation in Sect. 5 measures the second update's contribution. A Cholesky factorization is reused across updates. The refined map is unconstrained; fitting requires paired document vectors and linear solves, with no neural mapper training.

The norm factor also gives positive global scale equivariance. In exact arithmetic with nonzero predictions, $T(cB) = cT(B)$ for a common $c > 0$. Linearity of the ridge solve therefore scales each update by c; by induction both accumulated maps scale by c, leaving normalized inference unchanged. This property concerns a common coefficient scale, not arbitrary per-document changes or guaranteed finite-step cosine improvement.

Table 1. ARC2-Global fitting and gallery construction. All document vectors supplied to fitting or replacement are unit normalized.

1 Form paired anchors X_A, Y_A and augmented design $\widetilde{X} = [X_A, \mathbf{1}]$.
2 Compute centered Procrustes and raw affine B_0 using Eqs. (2)–(3).
3 Factor $\widetilde{X}\widetilde{X}^\top + 10I_m$ once.
4 For $t = 0, 1$: compute $T^{(t)}$ from Eq. (4), solve Eq. (7), and accumulate B_{t+1}.
5 Store B_2; map documents with $\widehat{y}_i = \mathcal{N}([x_i, 1]B_2)$.
6 Replace rows in S by y_i; search with destination query vectors.

3.4 Affine Proxies and Mixed Retrieval

The two corrections collapse into the final coefficient matrix B_2. For each document, compute a proxy $\widehat{y}_i = \mathcal{N}([x_i, 1]B_2)$ and construct

$$g_i = \begin{cases} y_i, & i \in S, \\ \widehat{y}_i, & i \notin S. \end{cases} \tag{9}$$

Every anchor is replaced by its actual fitted destination vector. For a unit destination query q, rank documents by $qg_i^\top$. True and proxy vectors therefore compete in one score space. This competition makes mixed-gallery quality a distinct evaluation target: proxy approximation quality alone does not determine how replacement changes their relative scores. We use exact float32 inner products, sorting by descending score and then ascending document row index for ties; dataset-defined identical query/document IDs are excluded where specified. Table 1 summarizes the complete computation.

3.5 Computation and Storage

The final mapper has pd_t coefficients regardless of anchor count. For $384 \to 768$, this is 295,680 float32 values, or 1,182,720 bytes (1.13 MiB), including the intercept. This is coefficient storage; fitting objects, document galleries, and encoder weights are additional. Application requires one affine multiplication and normalization per document, with $O(Npd_t)$ arithmetic for a complete proxy gallery.

Fitting cost depends on the solver backend. Beyond the cross-covariance and rectangular SVD, costing $O(md_sd_t)$ and $O(d_sd_t \min(d_s, d_t))$, respectively, the reference dual implementation forms an $m \times m$ Gram matrix in $O(m^2p)$ time, factors it in $O(m^3)$, and spends $O(m^2d_t + mpd_t)$ per correction. Its Gram storage is $O(m^2)$. A separately identified primal backend solves $(\widetilde{X}^\top\widetilde{X} + \lambda I_p)\Delta_t = \widetilde{X}^\top T^{(t)}$ instead, with $O(mp^2 + p^3)$ construction and factorization and $O(p^2)$ Gram storage; it supports larger paired sets. Each primal correction costs $O(mpd_t + p^2d_t)$. The equations are equivalent in exact arithmetic, while floating-point rankings can differ near ties. Updates use float64 and the final coefficient is stored as float32. Finally, $|A \cup S|$ measures logical destination-vector exposure in cached-vector evaluations. It does not measure physical encoding work, deployment latency, or end-to-end migration savings.

4 Experimental Setup

We ask three questions: does angular refinement improve fixed-supervision alignment, how does it compare with other mappers at a matched encoding budget, and which gains transfer to a prospectively held collection? All experiments use exact retrieval, isolating representation quality from approximate-index effects.

Collections and migrations. Development uses FiQA (57,638 documents, 500 development queries), NanoNQ (5,035/50), NanoHotpotQA (5,090/50), and NanoFEVER (4,996/50), drawn from BEIR and NanoBEIR [10,18]. FiQA informed method selection after its scores were inspected; all four collections are development evidence. We subsequently fixed a separate confirmation on the BEIR release of TREC-COVID [12,10]. Its input manifest contains all 171,332 documents and 50 queries. For this study, labels remained closed until every ranking and endpoint had been sealed and separately verified. This distinguishes development evidence from the prospective test. We evaluate two 384-to-768-dimensional migrations: `all-MiniLM-L6-v2` to `all-distilroberta-v1`, and `e5-small-v2` to `e5-base-v2`.[1] Model revisions and input hashes are pinned. Text encoding uses normalized float32 vectors, a maximum of 128 tokens, and the E5 query/passage prefixes.

Matched supervision and exposure. The main matrix crosses four development collections, two migrations, three paid-document budgets (25/50/75%), six methods, and seeds 17, 29, 41, 53, and 67. All methods fit the same ordered 1,024 paired anchors and replace the same paid document rows. Paid sets contain all anchors and are nested, with exactly $\lfloor \rho N \rfloor$ unique target document vectors; the accounting is $|A \cup S|$, including anchors. On the Nano collections and TREC-COVID, independent seeded streams sample anchors without replacement and a document permutation; unseen permutation rows fill the budget. FiQA retains its historical anchor/order arrays. Neither fitting nor selection uses relevance labels. Paid nonanchor vectors provide replacement only in this fixed-anchor regime. A separate FiQA experiment holds the 50% paid set fixed and compares fitting on 1,024 versus all 28,819 paid pairs. Development fixed-anchor ARC2 rankings use the float64 dual backend; all-paid and TREC-COVID fits use the float64 primal backend in Sect. 3.5. The all-paid study reuses the dual fixed-anchor comparisons. Both backends implement the same two-update rule.

Comparison methods. We retain the full five-baseline panel: centered rectangular Procrustes (an A2M adaptation); LA2M with 20 source-anchor clusters, local alignment for clusters with at least two anchors and global fallback for singletons; SimpleLinear; affine ridge with $\lambda = 10$ and a regularized bias; and EmbeddingConverter [16,2,17]. SimpleLinear uses pinned official code: despite its name, the locked configuration is a five-layer SELU MLP, trained for the final 50 epochs with batch size 4,028 and the MSE plus cosine loss, without qrel-based stopping.

[1] Pinned maintainer cards: MiniLM, DistilRoBERTa, E5-small, and E5-base.

Converter is a paper-specification adaptation with a 50,000-update selection run, batch size 64, and local neighborhood size 100. Paired-vector validation MAE selects the earliest minimum at 250-update checkpoints; a fresh fit uses all fitting pairs for that selected length. Unlike the original retrieval-based validation, this adaptation uses paired-vector MAE to respect paired-document-only supervision. These are the declared implementations and training policies, rather than claims to reproduce every original-paper setting. ARC2-Global remains fixed at two tangent updates and $\lambda = 10$ throughout.

Retrieval and inference. Destination queries search a single gallery containing unit-normalized proxies and paid true vectors, using exact float32 inner products and top-100 retrieval. Score ties use original document row order; identical query/document IDs are excluded. Native Full-X and Full-Y use their own query/document encoders throughout and provide reference operating points. Our primary metric is graded nDCG@10 with gain $2^{\text{rel}} - 1$; auxiliary Recall@100 measures coverage over all positive judgments, including positives outside a Nano corpus. All queries count; unjudged documents are nonrelevant. The prospective figure reports both metrics; a companion Recall table is included in the manuscript-source package.

Each locked analysis uses 20,000 paired query-bootstrap draws after averaging the five mapping seeds. Development macro effects weight the 24 collection/migration/budget conditions equally and resample queries within each collection. Within each analysis, the same sampled query indices are shared across methods, budgets and migrations for that collection. Saved 99% percentile intervals give approximate 95% coverage for the five-comparison family; the ablations have a separate five-contrast family. The all-paid experiment uses its predeclared 32-contrast family. TREC-COVID has 30 primary contrasts (five baselines, two migrations, three budgets), with approximately 99.8333% individual percentile intervals for approximate 95% family coverage. Its independent units are the 50 queries. Recall intervals are nominal 95% descriptive intervals. An interval touching zero is unresolved. These intervals condition on the five observed mapping seeds and fixed collections and encoder pairs; development intervals do not adjust for method selection.[2]

5 Results

We evaluate ranking quality, the value of angular refinement, transfer to a held collection, and useful operating points under limited document exposure.

5.1 Broad Retrieval Gains with a Compact Adapter

ARC2-Global achieves the highest macro nDCG@10 among the six evaluated methods (Table 2). Its gain over Procrustes is 0.0243, with family-adjusted interval $[0.0181, 0.0308]$. The effect extends across both migrations: descriptive

[2] Code, full baseline settings (`configs/baselines.json`) and manifests are packaged in a separate research reproducibility artifact. An anonymous public link is pending.

Table 2. Development means at matched 1,024-pair supervision, equally weighting 24 conditions. Differences are ARC2 minus baseline; intervals give approximate 95% family coverage. Bold/underline mark the best/second unrounded adapter means; blue identifies ARC2.

Method	nDCG@10 ↑	R@100 ↑	Δ nDCG	Family interval
ARC2-Global	**0.5275**	<u>0.8659</u>	—	—
Procrustes	<u>0.5032</u>	**0.8661**	+0.0243	[+0.0181, +0.0308]
LA2M20	0.4234	0.7758	+0.1041	[+0.0861, +0.1229]
SimpleLinear	0.3554	0.6130	+0.1721	[+0.1499, +0.1947]
Affine ridge	0.4318	0.7441	+0.0957	[+0.0754, +0.1164]
Converter	0.3663	0.7641	+0.1612	[+0.1427, +0.1799]

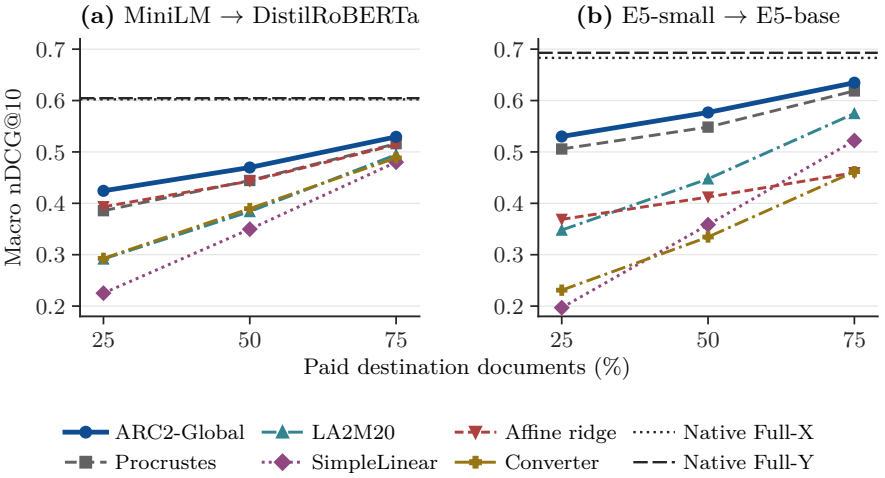


Fig. 2. Development quality at anchor-inclusive paid-document coverage. Points average five seeds and then weight the four collection means equally. All six adapters and both native references are retained. Lines connect the three measured budgets; they do not imply intermediate measurements.

pair-wise gains are 0.0257 for MiniLM and 0.0229 for E5. Each of the four collection-level macros also favors ARC2 over every baseline. At the finer condition level, ARC2 leads Procrustes in 23 of 24 point means and every other baseline in all 24. Figure 2 shows the complete budget curves.

The improvement concerns top-ranked relevance: macro Recall is 0.8659 versus Procrustes' 0.8661, with descriptive difference interval $[-0.0049, 0.0043]$. The single negative Procrustes nDCG point difference is FiQA/E5 at 75% (-0.0036). Native Full-X and Full-Y provide the full-corpus operating context.

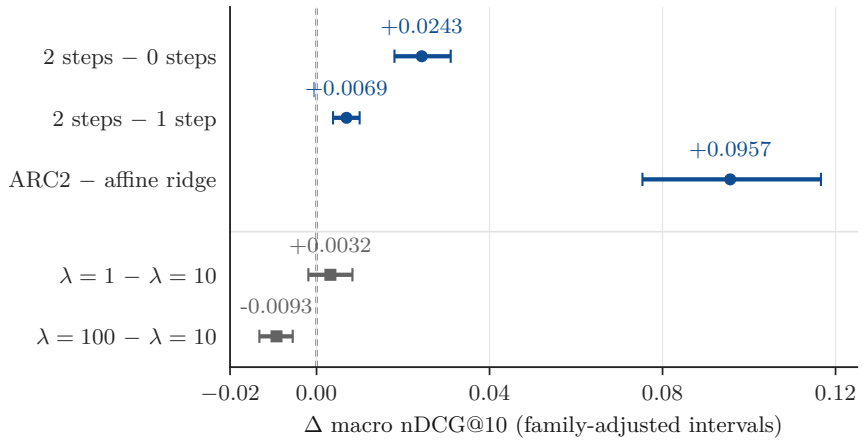


Fig. 3. Development mechanism controls. Dots show equal-condition macro nDCG differences; bars retain the saved 99% intervals of the separate five-contrast family. The last two rows change the penalty at two steps. All intervals use the same continuous horizontal scale.

5.2 The Second Correction Adds Value without an Inference Stage

A second correction adds 0.0069 macro nDCG beyond one step, with family interval $[0.0038, 0.0100]$. The point gain appears in 22 of 24 conditions and in each of the five seed-level macros. Recomputing the tangent target therefore provides a repeatable development benefit while the final mapper remains one affine stage. The complete initialized refinement gains 0.0957 over affine ridge (Fig. 3); this comparison jointly evaluates tangent targeting and the retained alignment prior.

The penalty controls favor moderate regularization. Changing λ from 10 to 1 gives $+0.0032$, with an interval crossing zero; changing it to 100 gives -0.0093, with a wholly negative interval. The evaluated method keeps $\lambda = 10$ and exactly two corrections.

5.3 Prospective Transfer: Strong E5 Gains and MiniLM Coverage

On the held collection, E5 ARC2 exceeds the evaluated Converter and affine ridge at **all three migration budgets**. At 25% coverage, the Converter gain is 0.2572, with family interval $[0.1757, 0.3376]$. At 50%, ARC2 reaches 0.5904 nDCG versus Converter's 0.3590 and ridge's 0.3281 (Table 3). Across both migrations, the 30-contrast family supports five improvements over Converter, four over ridge and two over SimpleLinear. The table marks all eleven supported comparisons.

MiniLM provides a useful partial-migration operating point: at 75% coverage, ARC2 reaches 0.5195 nDCG versus native Full-Y's 0.5203, while one quarter of the gallery still uses proxies. Its Recall@100 point mean is the highest among

Table 3. Prospective TREC-COVID nDCG@10 (higher is better), averaged over five seeds. †: positive ARC2-minus-baseline interval in the 30-contrast family. Unmarked baseline cells are unresolved. Bold/underline rank the six adapters before rounding; native references are separate.

Method	MiniLM → DistilRoBERTa			E5-small → E5-base		
	25%	50%	75%	25%	50%	75%
ARC2-Global	0.4619	0.5000	<u>0.5195</u>	<u>0.5628</u>	<u>0.5904</u>	0.6098
Procrustes	<u>0.4620</u>	0.5046	**0.5195**	**0.5660**	**0.6067**	**0.6198**
LA2M20	**0.4634**	<u>0.5073</u>	0.5186	0.5206	0.5898	<u>0.6188</u>
SimpleLinear	0.4610	**0.5076**	0.5193	0.4299†	0.5287†	0.5785
Affine ridge	0.4322	0.4854	0.5012†	0.2986†	0.3281†	0.3278†
Converter	0.4224†	0.4780†	0.5033	0.3056†	0.3590†	0.4098†
Native Full-X	0.4935 (native)			0.6805 (native)		
Native Full-Y	0.5203 (native)			0.6421 (native)		

Table 4. E5 nDCG: ARC2 at 25% exceeds Converter and ridge at 75% on all five collections. Blue identifies ARC2. Proc.: Procrustes; SL: SimpleLinear; native references show full-corpus quality. TREC-COVID is prospective; other rows are development. Cross-budget point comparisons are post-hoc descriptive.

Collection	25% coverage			75% coverage			Native	
	ARC2	Proc.	LA2M20	SL	Ridge	Conv.	Full-X	Full-Y
FiQA	0.2731	0.2718	0.1416	0.2781	0.1610	0.2095	0.3465	0.3891
NanoNQ	0.5071	0.4852	0.3640	0.4839	0.4428	0.4021	0.6595	0.6669
NanoHotpotQA	0.5845	0.5512	0.3869	0.6174	0.5053	0.5016	0.8278	0.8193
NanoFEVER	0.7555	0.7147	0.4991	0.7087	0.7290	0.7326	0.8984	0.8963
TREC-COVID	0.5628	0.5660	0.5206	0.5785	0.3278	0.4098	0.6805	0.6421

the six adapters at each budget, reaching 0.0998 at 75% versus native Full-Y's 0.1052. E5 reaches 0.0997 Recall at 75%, versus Full-Y's 0.1162 (Fig. 4). These auxiliary results distinguish coverage at 100 from relevance near the top; proximity in nDCG is not an equivalence test.

Geometric alignment remains competitive: before rounding, Procrustes has higher nDCG point means in all six cells, although its MiniLM/75% difference is below 0.0001. All Procrustes and LA2M nDCG family intervals cross zero. The confirmed advantages therefore concern the evaluated Converter, ridge and SimpleLinear configurations.

5.4 Higher E5 Quality at Lower Document Exposure

A useful budget pattern extends across all five collections (Table 4): E5 ARC2 at 25% coverage exceeds both Converter and affine ridge at 75%. On TREC-COVID, the respective nDCG means are 0.5628, 0.4098 and 0.3278. The advan-

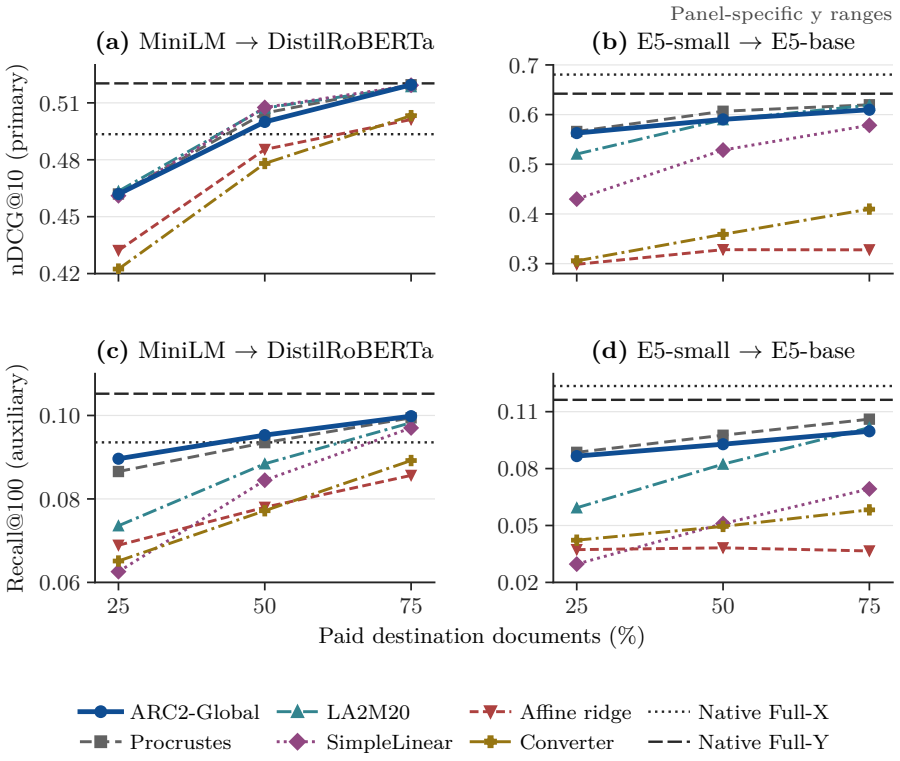


Fig. 4. Prospective TREC-COVID at fixed 1,024-anchor supervision: (a,b) primary nDCG@10; (c,d) auxiliary Recall@100. Points average five seeds. All six adapters and native references are shown, with explicitly panel-specific vertical ranges. Recall curves are descriptive.

tage is visible even after the two comparators receive a much larger set of true destination vectors.

This is a post-hoc descriptive comparison of the measured budgets, rather than an additional member of the matched-budget inferential family. The table retains 25% geometric alternatives, 75% SimpleLinear and both native references. It identifies observed quality–exposure operating points; physical savings and the minimum budget needed for equal quality require separate measurements.

5.5 Paid Pairs Strengthen MiniLM Adaptation

Using already-paid pairs improves MiniLM adaptation without increasing the 50% document budget. ARC2 rises from 0.3075 to 0.3671 nDCG, a gain of 0.0596 with family interval $[0.0393, 0.0806]$. In matched all-paid comparisons it exceeds

Table 5. FiQA at 50% coverage. Fixed/All fit 1,024/28,819 pairs; N/R denote nDCG@10/Recall@100. On All-N baseline cells, †/‡ mark positive/negative ARC2 differences in the 32-contrast family; unmarked comparisons are unresolved. Bold/underline rank matched adapter columns; native rows are references.

Method	MiniLM → DistilRoBERTa			E5-small → E5-base		
	Fixed N	All N	All R	Fixed N	All N	All R
ARC2-Global	**0.3075**	<u>0.3671</u>	<u>0.6922</u>	**0.3116**	0.2919	0.5595
Procrustes	0.2858	0.3334†	0.6905	<u>0.3100</u>	**0.3362**‡	**0.6534**
LA2M20	0.2377	0.3389†	**0.6950**	0.2300	<u>0.3293</u>‡	<u>0.6473</u>
SimpleLinear	0.2360	0.3323†	0.6778	0.1942	0.1679†	0.3618
Affine ridge	<u>0.2890</u>	**0.3673**	0.6800	0.1479	0.2270†	0.4245
Converter	0.2496	0.3258†	0.6633	0.1397	0.2625	0.5789
Native Full-X	0.3900 (native)		0.7002	0.3465 (native)		0.6453
Native Full-Y	0.4047 (native)		0.7039	0.3891 (native)		0.6968

Procrustes, LA2M, SimpleLinear and Converter; ridge is unresolved. Table 5 makes these family decisions visible.

E5 changes the method ordering: ARC2 trails all-paid Procrustes and LA2M, exceeds SimpleLinear and ridge, and is unresolved against Converter. Its fixed-to-all-paid change, 0.3116 to 0.2919, is unresolved. Fixed-anchor MiniLM ARC2 also trails every all-paid baseline in the cross-regime family comparisons. All-paid fitting retains $\lambda = 10$ in the summed objective; it tests the fixed procedure at larger supervision, rather than isolating data quantity at a fixed effective penalty.

Evaluate the gallery that will be searched. With each fitted map fixed, true-vector replacement raises MiniLM all-paid nDCG from 0.3510 to 0.3671, but lowers E5 from 0.3149 to 0.2919. For E5, Recall falls from 0.6542 to 0.5595 and the paid-document share of top-10 results falls from 0.4916 to 0.0647, despite 50% paid coverage. These measured replacement effects make mixed-gallery relevance the appropriate selection criterion: proxy-only quality can favor a different operating point from the gallery containing native and mapped vectors.

6 Practical Value of Directional Adaptation

Compact inference with stronger development ranking. The 295,680 float32 coefficients occupy 1.13 MiB, or 114.7 times fewer parameter bytes than the evaluated Converter. Both corrections are absorbed into this matrix. Affine ridge has the same coefficient count; refinement provides the measured quality benefit, while affine mapping provides compactness.

When the adapter is useful. ARC2 targets required destination-model queries during staged document replacement. E5 demonstrates useful quality at limited

exposure; MiniLM benefits from fitting all paid pairs without increasing document coverage. Procrustes and LA2M remain competitive on the held collection, so deployment choice should follow encoder-specific mixed-gallery validation. When source queries remain acceptable, Full-X is also viable: all 24 partial-development five-seed ARC2 condition means are below it.

Efficiency in the measured units. Destination-vector exposure, coefficient storage and the absence of neural mapper training describe the demonstrated resource advantages. Physical encoding time, index resources and serving SLOs require deployment measurements.

7 Conclusion

ARC2-Global turns two tangent ridge corrections into one compact affine adapter. It leads five alternatives in development macro nDCG, transfers substantial E5 gains over Converter and ridge to TREC-COVID, and exposes useful 25%-versus-75% quality–coverage operating points. Stronger all-paid MiniLM adaptation and measured replacement effects further show why migration should be evaluated in the mixed gallery that serves destination queries. Angular refinement offers a practical route to budgeted index migration.

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